

DENIS POLGUL

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Data Analyst with a Master's in Data Science and hands-on experience in statistical modelling, machine learning, and deep learning. Aptitude for leveraging data analysis to extract insights, facilitate decision-making, and enhance strategic initiatives. Adept at tailoring predictive models, automating workflows, and visualizing data to support business intelligence. Ability to determine trends and facilitate predictive analytics for business insights coupled with orchestrating strategic planning processes.

SKILLS & COMPETENCIES

Data Mining | ETL | Machine Learning Basics | Data Modeling | Business Intelligence | Data Analysis | Problem-solving Skills | Post-hoc Analysis | Data Interpretation | Statistical Analysis | Data Cleaning & Visualization | Data Wrangling | A/B Testing | Communication Skills | Continuous Process Improvements | Strategic Planning | Predictive Modeling | Random Forests

EDUCATION

Master's in Data Science | *University of Colorado Boulder, Boulder, CO* (2023)

Coursework: Statistical Modeling | Linear Regression | Statistics | Databases | Machine Learning

TECHNICAL SKILLS

Data Analysis & Statistics: Descriptive Statistics | Hypothesis Testing | Data Interpretation)

Data Manipulation & Cleaning: Excel (Pivot tables, VLOOKUP, Data Validation) | Python (NumPy and Pandas for Data Wrangling) | Handling missing data, Outlier Detection, Normalization

Programming: R | SQL | Python (Jupyter Notebooks and Google Colab)

Data Visualization: Dashboards in Watson Studio | Charts from Python Libraries (Matplotlib, Seaborn) | ggplot2

Other Tools: MS Suite (Excel, Word) & Google Suite for collaboration and Post Hoc Analysis | Jupyter Notebook and Google Colab for Data Analysis

PROFESSIONAL EXPERIENCE

Data Analyst Intern | *Derive Systems, Denver, CO*

05/2022 – 12/2022

- Facilitated informed decision-making and strategic planning by creating presentations for linear models of fuel savings estimation, providing data-driven insights for potential clients.
- Decreased manual workload, improved efficiency, and automated a manual Excel process into a Python script that generated client-ready PowerPoint presentations from vehicle data.
- Identified key trends and supported predictive analytics for business insights by cleaning data, performing statistical analysis, and developing models using R and Python.

KEY PROJECTS

Deep Learning – Kaggle's Histopathologic Cancer Detection | [GitHub](#)

- Classified histopathological images and enhanced the accuracy of tissue identification for medical diagnostics by developing a Convolutional Neural Network (CNN) utilising TensorFlow and Keras.
- Increased generalization and prevented overfitting by performing hyperparameter tuning and employing various techniques, including early stopping, elevating model performance.
- Boosted diagnostic reliability and aided in early detection by leveraging preprocessing techniques, attaining 85% precision in distinguishing malignant from benign cell tissues.

Deep Learning Project – Road Sign Image Classification | [GitHub](#)

- Classified road signs across 43 categories by developing a Convolutional Neural Network (CNN) classification model, guaranteeing precise identification of traffic signs.
- Enhanced model generalization against varying real-world conditions by pre-processing image data while training neural networks and assessing model performance.

ADDITIONAL EXPERIENCE

Graduate Teaching Assistant | *University of Colorado, Boulder*

- Worked as a teaching assistant for five semesters (2.5 years), supporting two courses each semester. Helped students understand key statistical principles, including probability, estimation techniques, and hypothesis testing methods, explaining fundamental concepts.
- Supported students in gaining familiarity with regression analysis techniques, covering linear regression models, ANOVA, and generalized linear models (GLMs).